

Road Transportation Accident

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ABSTRACT

This project, 'Road Transportation Accident ' is about preparing a real-world road transportation accident dataset for creating meaningful accident insights and dashboards. The project focuses on understanding accident-related information such as drivers, vehicles, weather conditions, road surfaces, junction types, and casualty details. It involves inspecting the dataset, identifying duplicate and missing values, and applying different data-cleaning techniques to make the data accurate, consistent, and ready for visualization and reporting. The final outcome of the project is a clean and structured transportation accident dataset that can be used to monitor accident patterns, support road safety reporting, and provide useful insights for decision-making.

DATASET OVERVIEW

The dataset contains **12,316 rows and 32 columns**. The columns included are:

```
# column names  
df.columns
```

```
Index(['Time', 'Day_of_week', 'Age_band_of_driver', 'Sex_of_driver',  
      'Educational_level', 'Vehicle_driver_relation', 'Driving_experience',  
      'Type_of_vehicle', 'Owner_of_vehicle', 'Service_year_of_vehicle',  
      'Defect_of_vehicle', 'Area_accident_occured', 'Lanes_or_Medians',  
      'Road_allignment', 'Types_of_Junction', 'Road_surface_type',  
      'Road_surface_conditions', 'Light_conditions', 'Weather_conditions',  
      'Type_of_collision', 'Number_of_vehicles_involved',  
      'Number_of_casualties', 'Vehicle_movement', 'Casualty_class',  
      'Sex_of_casualty', 'Age_band_of_casualty', 'Casualty_severity',  
      'Work_of_casualty', 'Fitness_of_casualty', 'Pedestrian_movement',  
      'Cause_of_accident', 'Accident_severity'],  
      dtype='object')
```

These columns provide information about **drivers, vehicles, road conditions, weather, collisions, casualties, and accident severity**, which helps in understanding road

transportation accident patterns and preparing the data for dashboard and reporting purposes.

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	AA	AB	AC	AD	AE	AF	AG						
1	Time	Day of Week	Age	Sex	of Vehicle	Education	Vehicle	Driving	Type of Owner	Service	Defect	c Area	so Lanes	o Road	all Types	o Road	su Road	Light	co Weather	Type of	Number	Number	Vehicle	Casualty	Sex	of	Age	ban	Casualty	Work	of Fitness	Pedestri	Cause	c Accident	severity				
2	17:02:00	Monday	18-30	Male	Above hi	Employe	1-2yr	Automol	Owner	Above K	No defec	Residential	areas	Tangent	No junoti	Asphaltr	Dry	Daylight	Normal	Collision	2	2	Going st	na	na	na	na							Not a Pe	Moving t	Slight Injury			
3	17:02:00	Monday	31-50	Male	Junior hi	Employe	Above K	Public (	Owner	5-10yrs	No defec	Office an	Undivide	Tangent	No junoti	Asphaltr	Dry	Daylight	Normal	Vehicle v	2	2	Going st	na	na	na	na							Not a Pe	Overtaki	Slight Injury			
4	17:02:00	Monday	18-30	Male	Junior hi	Employe	1-2yr	Lorry (H	Owner		No defec	Recreat	other		No junoti	Asphaltr	Dry	Daylight	Normal	Collision	2	2	Going st	Driver or	Male	31-50		3	Driver						Not a Pe	Changin	Serious Injury		
5	01:00:00	Sunday	18-30	Male	Junior hi	Employe	5-10yr	Public (	Governmental		No defec	Office an	other	Tangent	Y Shape	Earthros	Dry	Darknes	Normal	Vehicle v	2	2	Going st	Pedestri	Female	18-30		3	Driver	Normal					Not a Pe	Changin	Slight Injury		
6	01:00:00	Sunday	18-30	Male	Junior hi	Employe	2-5yr		Owner	5-10yrs	No defec	Industria	other		Tangent	Y Shape	Asphaltr	Dry	Darknes	Normal	Vehicle v	2	2	Going st	na	na	na	na							Not a Pe	Overtaki	Slight Injury		
7	14:15:00	Friday	31-50	Male		Unknown									Y Shape		Dry	Daylight	Normal	Vehicle v	1	1	U-Turn	Driver or	Male	31-50		3	Driver	Normal					Not a Pe	Overloac	Slight Injury		
8	17:30:00	Viednes	18-30	Male	Junior hi	Employe	2-5yr	Automol	Owner		No defec	Resident	Undivide	Tangent	Crossing		Dry	Daylight	Normal	Vehicle v	1	1	Moving t	Driver or	Female	18-30		3	Driver	Normal					Not a Pe	Other	Slight Injury		
9	17:20:00	Friday	18-30	Male	Junior hi	Employe	2-5yr	Automol	Governm	Above K	No defec	Resident	other	Tangent	Y Shape	Asphaltr	Dry	Daylight	Normal	Vehicle v	2	1	U-Turn	na	na	na	na							Normal		Not a Pe	No priori	Slight Injury	
10	17:20:00	Friday	18-30	Male	Junior hi	Employe	Above K	Lorry (H	Owner	1-2yr	No defec	Industria	other	Tangent	Y Shape	Earthros	Dry	Daylight	Normal	Collision	2	1	Going st	Pedestri	Male	Under 18		3	Driver	Normal					Crossing	Changin	Slight Injury		
11	17:20:00	Friday	18-30	Male	Junior hi	Employe	1-2yr	Automol	Owner	2-5yrs	No defec	Resident	Undivide	Tangent	Y Shape	Asphaltr	Dry	Daylight	Normal	Collision	2	1	U-Turn	Passeng	Male	18-30		3	Driver	Normal						Not a Pe	Moving t	Serious Injury	
12	14:40:00	Saturday	18-30	Male	Above hi	Owner	1-2yr	Public (	Owner	Unknown	No defec	Resident	other	Tangent	No junoti	Asphaltr	Dry	Daylight	Normal	Collision	2	1	Turnove	na	na	na	na							Normal		Not a Pe	Changin	Serious Injury	
13	14:40:00	Saturday	31-50	Male	Above hi	Employe	No Lisen	Automol	Owner		No defec	Office an	Undivide	Tangent	No junoti	Earthros	Dry	Daylight	Normal	Collision	2	1	Going st	Driver or	Male	18-30		3	Driver	Normal						Not a Pe	No priori	Serious Injury	
14	17:45:00	Thursday	18-30	Male	Junior hi	Employe	1-2yr	Public (	Owner	2-5yrs	No defec	Office an	Double c	Escapm	No junoti	Asphaltr	Dry	Daylight	Normal	Collision	2	2	Going st	na	na	na	na							Driver	Normal		Not a Pe	No distat	Slight Injury
15	17:45:00	Thursday	31-50	Male	Junior hi	Employe	5-10yr	Lorry (H	Owner	Above K	No defec	Office an	other	Tangent	No junoti	Asphaltr	Dry	Daylight	Normal	Collision	2	2	Valting h	na	na	na	na							Other	Normal		Not a Pe	No priori	Slight Injury
16	17:45:00	Thursday	31-50	Male	Junior hi	Employe	Above K	Automol	Owner	1-2yr	No defec	Office an	Undivide	Escapm	No junoti	Asphaltr	Dry	Daylight	Normal	Collision	2	2	Going st	Driver or	Female	18-30		3	Driver	Normal							Not a Pe	No distat	Serious Injury
17	22:45:00	Monday	18-30	Female	Junior hi	Employe	Above K	Lorry (H	Owner		No defec	Office an	One way	Tangent	Y Shape	Asphaltr	Vet or d	Darknes	Raining	Collision	2	3	Going st	na	na	na	na									Not a Pe	No distat	Serious Injury	
18	22:45:00	Monday	18-30	Male	Junior hi	Owner	5-10yr	Public (	Owner		No defec	Office an	Undivide	Tangent	Y Shape	Asphaltr	Vet or d	Darknes	Raining	Collision	2	3	Going st	na	na	na	na							Driver	Normal		Not a Pe	Moving t	Slight Injury
19	22:45:00	Monday	18-30	Male	Element	Employe	2-5yr	Public (	Owner		No defec	Office an	other	Tangent	Y Shape	Asphaltr	Vet or d	Darknes	Raining	Collision	2	3	Going st	na	na	na	na							Driver	Normal		Not a Pe	Changin	Serious Injury
20	22:45:00	Monday	18-30	Male	Junior hi	Employe	2-5yr		Owner	Above K	No defec	Resident	other	Tangent	Y Shape	Asphaltr	Vet or d	Darknes	Raining	Collision	2	3	Moving t	na	na	na	na							Unemplo	Normal		Not a Pe	No priori	Slight Injury
21	08:20:00	Tuesday	18-30	Male	Junior hi	Employe	Below 1yr	Long lorr	Owner	5-10yrs	No defec	Resident	One way	Tangent	Y Shape	Asphaltr	Dry	Daylight	Normal	Collision	2	1	Going st	Driver or	Male	Under 18		3	Driver	Normal						Not a Pe	Moving t	Serious Injury	
22	08:20:00	Tuesday	Under 18	Male	Junior hi	Employe	2-5yr	Lorry (H	Owner		No defec	Other	One way	Tangent	Y Shape	Asphaltr	Dry	Daylight	Normal	Collision	2	1	Going st	na	na	na	na							Driver	Normal		Not a Pe	Getting c	Slight Injury
23	15:10:00	Thursday	18-30	Male	Junior hi	Employe	2-5yr	Lorry (H	Owner	5-10yrs	No defec	Church	Undivide	Tangent	Crossing	Asphaltr	Dry	Daylight	Normal	Collision	1	1	Moving t	Driver or	Male	18-30		3	Driver	Normal							Not a Pe	Overtaki	Serious Injury
24	12:11:00	Thursday	18-30	Male	Element	Employe	5-10yr	Lorry (H	Owner		No defec	Recreat	other	Tangent	Y Shape	Asphaltr	Dry	Daylight	Normal	Other	2	1	Getting c	Passeng	Female	31-50		3	Driver	Normal							Not a Pe	Overloac	Serious Injury
25	12:11:00	Thursday	31-50	Male	Junior hi	Employe	Above K	Public (	Owner			Industria	Undivide	Tangent	Y Shape	Asphaltr	Dry	Daylight	Normal	Rollover	2	1	Reversin	Passeng	Male	18-30		3	Employe	Normal							Not a Pe	Changin	Serious Injury
26	18:36:00	Viednes	18-30	Male		Unknown						Resident	other		Crossing	Asphaltr	Dry	Daylight	Normal	Collision	2	3	Turnove	na	na	na	na							Driver	Normal		Not a Pe	Improper	Slight Injury
27	18:36:00	Viednes	18-30	Male	Junior hi	Employe	2-5yr	Taxi	Owner	Above K	No defec	Resident	other	Tangent	Crossing	Asphaltr	Dry	Daylight	Normal	Rollover	2	3	Going st	Driver or	Female	31-50		3	Self-emp	Normal							Not a Pe	Other	Slight Injury
28	18:36:00	Viednes	18-30	Unknown	Junior hi	Employe	5-10yr		Owner	2-5yrs	No defec	Office an	Double c	Tangent	Crossing	Asphaltr	Dry	Daylight	Normal	Collision	2	3	Going st	Pedestri	Male	Under 18		3	Driver	Normal					Crossing	Changin	Slight Injury		
29	14:35:00	Friday	18-30	Male	Above hi	Employe	2-5yr	Automol	Owner	2-5yrs	No defec	Market	Undivide	Tangent	No junoti	Asphaltr	Dry	Daylight	Normal	Collision	1	1	Going st	na	na	na	na							Driver	Normal		Not a Pe	Changin	Slight Injury
30	12:25:00	Tuesday	Over 51	Unknown		Unknown						Unknown	other	Tangent	No junoti	Asphaltr	Dry	Daylight	Normal	Vehicle v	2	1	Going st	Driver or	Male	31-50		3									Not a Pe	No priori	Slight Injury
31	12:25:00	Tuesday	18-30	Male		Employe	1-2yr	Lorry (H	Owner		No defec	Industria	other	Tangent	No junoti	Asphaltr	Dry	Daylight	Normal	Fall from	2	1	Going st	Driver or	Male	18-30		2	Driver	Normal							Not a Pe	Overspen	Slight Injury
32	15:30:00	Monday	18-30	Male	Junior hi	Employe	2-5yr	Automol	Owner		No defec	Resident	Two-way	Tangent	Crossing	Asphaltr	Dry	Daylight	Normal	Collision	1	1	Going st	na	na	na	na										Not a Pe	Changin	Slight Injury
33	18:10:00	Thursday	18-30	Male	Junior hi	Employe	Above K	Automol	Governm	Below 1yr	No defec	Resident	Undivide	Tangent	No junoti	Asphaltr	Dry	Daylight	Normal	Collision	1	1	Moving t	Driver or	Male	18-30		3	Driver								Not a Pe	No priori	Slight Injury

```
# shape
df.shape
```

ROWS - 12316

COLUMNS – 32

(12316, 32)

DATA CLEANING :

Duplicates:

The dataset was checked for duplicate records using the following command:

```
# duplicates  
df.duplicated().sum()
```

0

The output returned **0** ,indicating that there are no duplicate records in the dataset

Missing values Before Cleaning

```
# null values  
df.isnull().sum()
```

Time	0
Day_of_week	0
Age_band_of_driver	0
Sex_of_driver	0
Educational_level	741
Vehicle_driver_relation	579
Driving_experience	829
Type_of_vehicle	950
Owner_of_vehicle	482
Service_year_of_vehicle	3928
Defect_of_vehicle	4427
Area_accident_occured	239
Lanes_or_Medians	385
Road_allignment	142
Types_of_Junction	887
Road_surface_type	172
Road_surface_conditions	0
Light_conditions	0
Weather_conditions	0
Type_of_collision	155
Number_of_vehicles_involved	0
Number_of_casualties	0
Vehicle_movement	308
Casualty_class	0
Sex_of_casualty	0
Age_band_of_casualty	0
Casualty_severity	0
Work_of_casualty	3198
Fitness_of_casualty	2635
Pedestrian_movement	0
Cause_of_accident	0
Accident_severity	0

dtype: int64

Handled missing values:

- Educational_level → Filled using mode (most frequent value)
- Vehicle_driver_relation → Filled using mode
- Owner_of_vehicle → Filled using rule-based conditional filling based on Vehicle_driver_relation
- Driving_experience → Filled using group-based mode imputation based on Age_band_of_driver
- Road_surface_type → Filled using context-based imputation based on Weather_conditions
- Types_of_Junction → Filled using mapping + fallback conditional filling based on Area_accident_occured
- Vehicle_movement → Filled using a custom apply() function based on Type_of_collision
- Defect_of_vehicle → Replaced missing values with Unknown
- Work_of_casualty → Filled with Occupation not recorded
- Fitness_of_casualty → Filled with Unknown fitness status
- Type_of_vehicle → Filled using mode
- Service_year_of_vehicle → Filled using mode
- Area_accident_occured → Filled using mode
- Lanes_or_Medians → Filled using mode
- Road_alignment → Filled using mode
- Type_of_collision → Filled using mode

Missing values After Cleaning

```
df.isnull().sum()
Time 0
Day_of_week 0
Age_band_of_driver 0
Sex_of_driver 0
Educational_level 0
Vehicle_driver_relation 0
Driving_experience 0
Type_of_vehicle 0
Owner_of_vehicle 0
Service_year_of_vehicle 0
Defect_of_vehicle 0
Area_accident_occured 0
Lanes_or_Medians 0
Road_allignment 0
Types_of_Junction 0
Road_surface_type 0
Road_surface_conditions 0
Light_conditions 0
Weather_conditions 0
Type_of_collision 0
Number_of_vehicles_involved 0
Number_of_casualties 0
Vehicle_movement 0
Casualty_class 0
Sex_of_casualty 0
Age_band_of_casualty 0
Casualty_severity 0
Work_of_casualty 0
Fitness_of_casualty 0
Pedestrian_movement 0
Cause_of_accident 0
Accident_severity 0
dtype: int64
```